

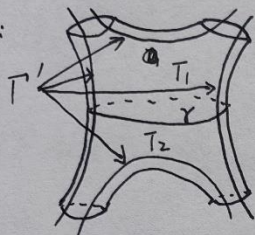
7.6. Fenchel-Nielsen Coordinates on Teichmüller Space

①

FN coordinate: length + twist

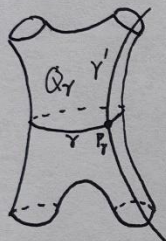
Length: S is cpt finite type, choose a trouser decomposition by maximal multicurve $\Gamma = \{\gamma_1, \dots, \gamma_{3g-3}\}$.
 $\forall \gamma \in \Gamma$ on S , $\varphi: S \rightarrow X$ representing a point $\tau \in T_S$, $\exists!$ geodesic on X in the homotopy class of $\varphi(\gamma)$ on X (for hyperbolic metric on X). $\xrightarrow{\text{determine}}$ Denote the length of this geodesic by $L_\gamma(\tau)$, define a map $FN_L: T_S \rightarrow (R_+^*)^\Gamma$ by $FN_L(\tau) = (L_\gamma(\tau))_{\gamma \in \Gamma}$.
 topological curves/trousers on $S \xrightarrow{\varphi} \text{hyperbolic geodesics/trousers on } X$.

Twist:



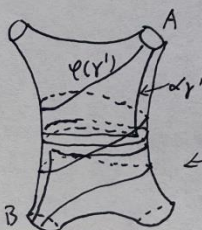
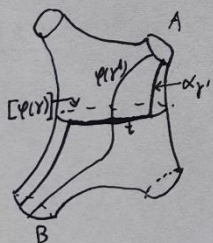
arcs join boundary components of each trouser.

Let Q_γ be the union of the (one or two) trousers with γ as one boundary component



$\gamma' \subset Q_\gamma$ of one of the curves of Γ' joining one point of the boundary of Q_γ to another, intersects γ once at p_γ .

$\varphi(\gamma), \gamma \in \Gamma$ cut $X = \varphi(S)$ into hyperbolic trousers, corresponding to the trousers of S . We get $[\varphi(\Gamma)], [\varphi(Q_\gamma)], [\varphi(\gamma)]$ $\xrightarrow{\text{homotopy class of } \varphi(\gamma)}$



$\exists! \alpha_\gamma$ homotopic to $[\varphi(\gamma')]$, joining boundary points of $[\varphi(Q_\gamma)]$, and consist 1. minimal geodesic arc joins $[\varphi(\gamma)]$ to the component of $\partial[\varphi(Q_\gamma)]$ containing one end of $\varphi(\gamma')$. 2. an immersed geodesic δ_γ , whose image lies in $[\varphi(\gamma)]$, denote length of δ_γ by $t_\gamma(\tau)$, positive if turning was right, negative if turning is left. 3. minimal geodesic joins $[\varphi(\gamma)]$ to component of $\partial[\varphi(Q_\gamma)]$ containing other end of $\varphi(\gamma')$.

Def: The twist coordinate map $FN_T: T_S \rightarrow R^\Gamma$ associates to $\tau \in T_S$ the vector of lengths $(t_\gamma(\tau))_{\gamma \in \Gamma}$

$\leftarrow$ around twice, length of δ_γ is $t + 2L_\gamma(\tau)$

Now, define a map $FN := (FN_L, FN_T) : T_S \rightarrow (R_+^*)^r \times R^n$ (2)

Theorem 7.6.3. (Fenchel - Nielsen coords) The map $FN : T_S \rightarrow (R_+^*)^r \times R^n$ is a homeomorphism.

Thm 7.6.4. Let S be a quasiconformal surface of finite type. The function $\ln L_\gamma : T_S \rightarrow R$ satisfies: $|\ln L_\gamma(T_1) - \ln L_\gamma(T_2)| \leq d(T_1, T_2)$ (Lipschitz continuous with constant 1)

Pf: $T_1, T_2 \in T_S$, by $\varphi_1 : S \rightarrow X_1, \varphi_2 : S \rightarrow X_2$.

Choose γ on S , γ generates $\langle \gamma \rangle \cong \mathbb{Z} \leq \pi_1(S, S)$. Consider covering $\pi_\gamma : \tilde{S}_\gamma \rightarrow S, (X_1)_\gamma \rightarrow X_1, (X_2)_\gamma \rightarrow X_2$

Those covering spaces are all annuli.

By 3.7. The modulus of a cylinder A of finite modulus is a non-Euclidean invariant: on a cylinder of modulus M , there is a unique simple closed geodesic for its hyperbolic geometry, and the hyperbolic length of this geodesic is π/M .

Thus, $M_1 = \frac{\pi}{L_\gamma(T_1)}, M_2 = \frac{\pi}{L_\gamma(T_2)}$. Let $f : X_1 \rightarrow X_2$ is quasiconformal map homotopic to $\varphi_2^{-1} \circ \varphi_1$, f maps $(\varphi_1)_*(\langle \gamma \rangle)$ to $(\varphi_2)_*(\langle \gamma \rangle)$ (f keeps homotopy type), so induces $\tilde{f}_\gamma : (X_1)_\gamma \rightarrow (X_2)_\gamma$. $\tilde{f}_\gamma$ is quasiconformal, $k(f) = k(\tilde{f}_\gamma)$.

By 4.3.2. (Grötzsch's thm) $f : A_m \rightarrow A_{m'}$ be k -quasiconformal homeomorphism, then $\frac{1}{k} \leq \frac{m}{m'} \leq k$.
 $d(T_1, T_2) := \inf_{\gamma \sim \varphi_2^{-1} \circ \varphi_1} \ln k(f) = \inf \ln k(\tilde{f}_\gamma) \geq \left| \ln \frac{M_2}{M_1} \right| = |\ln L_\gamma(T_1) - \ln L_\gamma(T_2)|$ □

Pro 7.6.5. The function FN_T is continuous.

Pf: Consider $T_1, T_2 \in T_S$, which is close $\tilde{T}_1 : \varphi : S \rightarrow X, \tilde{T}_2 : \varphi : S \rightarrow X_\mu, \mu \in M(X), \|\mu\|_\infty$ is small. Let $X = D/G_1, \mu$ induces $\tilde{\mu}$ on D .

Consider equation $\bar{\partial} f_{\tilde{\mu}} = \tilde{\mu} \partial f_{\tilde{\mu}}$ $\exists!$ quasiconformal map s.t. fix $1, -1, i$; when $\|\mu\|_\infty$ small $f_{\tilde{\mu}}$ is near id. Def $G_2 = f_{\tilde{\mu}} \circ G_1 \circ f_{\tilde{\mu}}^{-1}, X_\mu \cong D/G_2$.

Now lift the three geodesics defining $t_\gamma(T_1) : \tilde{A}_1$ of $A, \tilde{B}_1$ of $B, \tilde{\varphi}(\gamma_1)$ of $\varphi(\gamma_1)$, and end points are fixed point in G_1 , under $f_{\tilde{\mu}}$, they become ~~endpoints~~ fixed points in G_2 . Since $f_{\tilde{\mu}}$ is near id, $\tilde{A}_2 = f_{\tilde{\mu}}(\tilde{A}_1)$ near $\tilde{A}_1, \tilde{B}_2 = f_{\tilde{\mu}}(\tilde{B}_1)$ near $\tilde{B}_1, \tilde{\varphi}(\gamma_2) = f_{\tilde{\mu}}(\tilde{\varphi}(\gamma_1))$ near $\tilde{\varphi}(\gamma_1)$. Thus, end points change continuously, perpendiculars also, hence $\delta_{r_1}, \delta_{r_2}$, so $t_\gamma(T_2)$ is close to $t_\gamma(T_1)$ □
 (little disturb, little change) (hyperbolic element $\in \text{PSL}(2, R)$, has two fixed points, $f_{\tilde{\mu}}$ maps end points to end points)

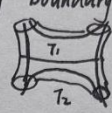
Proof of 7.6.3: We have already shown FN is continuous.

(3)

Firstly, we want to construct an inverse map. i.e. $\Psi: (R_+^*)^3 \times R^3 \rightarrow T_S$, s.t. $\Psi \circ FN = id$, $FN \circ \Psi = id$.

Given $l = (l_r)_{r \in \Gamma}$, by 3.5.8 (Given $a, b, c > 0$, $\exists$ trouser  length of A, B, C are a, b, c) we can construct hyperbolic trousers $\bar{T}_i$ corresponding to the trousers of S defined by Γ .

$\forall r \in \Gamma$, If r divides 2 different trousers $\bar{T}_1, \bar{T}_2$, let $r_1 \subset \partial \bar{T}_1$, $r_2 \subset \partial \bar{T}_2$ be boundary circles their length is l_r . In $\bar{T}_1, \bar{T}_2$ draw three geodesics joining boundary components, in $\bar{T}_1$ two of these curves intersect r_1 , in $\bar{T}_2$ two of them intersect r_2 .

 Choose one intersect r_1, r_2 get base points .

If r divides 1 trouser , it's same. Thus, we get $X_{l,t}$, $l = (l_r)_{r \in \Gamma}$, $t = (t_r)_{r \in \Gamma}$.

$\forall r \in \Gamma$, choose $A_r \subset S$, as an annulus, and disjoint. Let coordinate $\psi_r: R/\mathbb{Z} \times [-1, 1] \rightarrow A_r$ s.t.

r is $R/\mathbb{Z} \times \{0\}$, $r \cap A_r$ is $\{0\} \times [-1, 1]$, other curve is $\{\frac{1}{2}\} \times [-1, 1]$, base point p_r is origin.

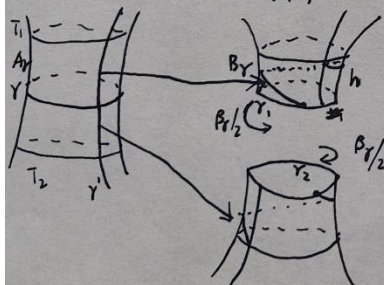
$\forall r$, by thm 3.8.3 (The collaring thm) Collar: r is closed geodesic, $A_\delta(r) = \{x \in X \mid d(x, r) < \delta\}$ And $A_{\eta(l_i)}(r_i)$ are collars, and they are disjoint. $\eta(l) = \frac{1}{2} \ln \frac{\cosh(\frac{l}{2}) + 1}{\cosh(\frac{l}{2}) - 1}$

$\zeta = \xi + i\eta$ $\xi \in R/l_r\mathbb{Z}$, $-h \leq \eta \leq h$ ζ is coordinates. We get $B_r \subset X_{l,t}$.

$\forall$ trouser T on S , let $T' := T - \bigcup_{r \in \Gamma} A_r$, trousers "shrink". Remove B_r , get T'' .

Choose a homeomorphism $\varphi_T: T' \rightarrow T''$. φ_T maps component of ∂A_r in T' to component of ∂B_r in T'' , with speed l_r ; maps curve in T' in T' to arcs of minimizing geodesics connecting boundary component of trousers of $X_{l,t}$.

On A_r , $\varphi(x + iy) = \xi + i\eta$, when $y \geq 0$ $\xi + i\eta = [x + \frac{p_r}{2} \cdot \frac{h-y}{y}] + i \frac{y}{h}$, $y \leq 0$, $\xi + i\eta = [x - \frac{p_r}{2} \cdot \frac{h+y}{y}] + i \frac{y}{h}$



So if $t_r \rightarrow t_r + k \cdot l_r$, get same surface, but they are different in twist. $k := \frac{p_r - \tilde{p}_r}{l_r}$

With, we get $\varphi: S \rightarrow X$, so we get Ψ

And $FN \circ \Psi$, $\Psi \circ FN$ is id from the construction.

□